



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Register Description

Table 27: Driver ID = 1: Sequencer Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x001A	15:0	0x0000	seq_preview_0_hg (R/W)
	Statistics engine configuration in preview-enter state 0: Off 1: On		
R0x001B	15:0	0x0000	seq_preview_0_flash (R/W)
	15:8	X	Reserved
	7	0x0000	Reserved
	6:0	0x0000	seq_flash_config_0 Flash configuration in preview-enter state 0x0000: Off 0x0001: On 0x0002: Locked/Manual 0x0003: Auto evaluate
	Seq preview 0 flash		
R0x001C	15:0	0x0040	seq_preview_0_skipframe (R/W)
	15:8	X	Reserved
	7	0x0000	seq_preview_0_skipframe_turn_off_fen 0: Frame valid remains on during skipped frame. 1: Frame valid is off output during skipped frame.
	6	0x0001	seq_preview_0_skipframe_skip_state 0: No skip state 1: Skip state
	5	0x0000	seq_preview_0_skipframe_skip_led_on Skip first frame when LED is on 0: When LED is on, first frame is not skipped. 1: When LED is on, first frame is skipped.
	4:0	X	Reserved
	Seq preview 0 skipframe		
R0x001D	15:0	0x0003	seq_preview_1_ae (R/W)
	Auto exposure configuration in preview state 0x0000: Off 0x0001: Dynamic range tracking 0x0002: Brightness tracking 0x0003: Rules only (ignore clipping)		
R0x001E	15:0	0x0002	seq_preview_1_fd (R/W)
	Flicker detection configuration in preview state 0x0000: Off 0x0001: On 0x0002: Manual		
R0x001F	15:0	0x0003	seq_preview_1_awb (R/W)
	Auto white balance configuration in preview state 0: Off 1: On		
R0x0020	15:0	0x0003	seq_preview_1_hg (R/W)
	Statistics engine configuration in preview state 0: Off 1: On		



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Table 27: Driver ID = 1: Sequencer Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0021	15:0	0x0000	seq_preview_1_flash (R/W)
	15:8	X	Reserved
	7	0x0000	Reserved
	6:0	0x0000	seq_flash_config_1 Flash configuration in preview state 0x0000: Off 0x0001: On 0x0002: Locked/Manual 0x0003: Auto evaluate
	Seq preview 1 flash		
R0x0022	15:0	0x0000	seq_preview_1_skipframe (R/W)
	15:8	X	Reserved
	7	0x0000	seq_preview_1_skipframe_turn_off_fen 0: Frame valid remains on during skipped frame. 1: Frame valid is off output during skipped frame.
	6	0x0000	seq_preview_1_skipframe_skip_state 0: No skip state 1: Skip state
	5	0x0000	seq_preview_1_skipframe_skip_led_on Skip first frame when LED on 0: When LED is on, first frame is not skipped. 1: When LED is on, first frame is skipped.
	4:0	X	Reserved
Seq preview 1 skipframe			
R0x0023	15:0	0x0004	seq_preview_2_ae (R/W)
	Auto exposure configuration in preview-leave state 0x0000: Off 0x0001: Dynamic range tracking 0x0002: Brightness tracking 0x0003: Rules only (ignore clipping)		
R0x0024	15:0	0x0000	seq_preview_2_fd (R/W)
	Flicker detection configuration in preview-leave state 0x0000: Off 0x0001: On 0x0002: Manual		
R0x0025	15:0	0x0001	seq_preview_2_awb (R/W)
	Auto white balance configuration in preview-leave state 0: Off 1: On		
R0x0026	15:0	0x0001	seq_preview_2_hg (R/W)
	Statistics engine configuration in preview-leave state 0: Off 1: On		



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Table 27: Driver ID = 1: Sequencer Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0027	15:0	0x0000	seq_preview_2_flash (R/W)
	15:8	X	Reserved
	7	0x0000	Reserved
	6:0	0x0000	seq_flash_config_2 Flash configuration in preview-leave state 0x0000: Off 0x0001: On 0x0002: Locked/Manual 0x0003: Auto evaluate
	Seq preview 2 flash		
R0x0028	15:0	0x0010	seq_preview_2_skipframe (R/W)
	15:8	X	Reserved
	7	0x0000	seq_preview_2_skipframe_turn_off_fen 0: Frame valid remains on during skipped frame. 1: Frame valid is off output during skipped frame.
	6	0x0000	seq_preview_2_skipframe_skip_state 0: No skip state 1: Skip state
	5	0x0000	seq_preview_2_skipframe_skip_led_on Skip first frame when LED On 0: When LED is on, first frame is not skipped. 1: When LED is on, first frame is skipped.
	4	0x0001	Reserved
	3:0	X	Reserved
	Seq preview 2 skipframe		
R0x0029	15:0	0x0000	seq_preview_3_ae (R/W)
	Auto exposure configuration in capture-enter state 0x0000: Off 0x0001: Dynamic range tracking 0x0002: Brightness tracking 0x0003: Rules only (ignore clipping)		
R0x002A	15:0	0x0000	seq_preview_3_fd (R/W)
	Flicker detection configuration in capture-enter state 0x0000: Off 0x0001: On 0x0002: Manual		
R0x002B	15:0	0x0000	seq_preview_3_awb (R/W)
	Auto white balance configuration in capture-enter state 0: Off 1: On		
R0x002C	15:0	0x0000	seq_preview_3_hg (R/W)
	Statistics engine configuration in capture-enter state 0: Off 1: On		



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Table 27: Driver ID = 1: Sequencer Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x002D	15:0	0x0000	seq_preview_3_flash (R/W)
	15:8	X	Reserved
	7	0x0000	Reserved
	6:0	0x0000	seq_flash_config_3 Flash configuration in capture-enter state 0x0000: Off 0x0001: On 0x0002: Locked/Manual 0x0003: Auto evaluate
	Seq preview 3 flash		
R0x002E	15:0	0x0040	seq_preview_3_skipframe (R/W)
	15:8	X	Reserved
	7	0x0000	seq_preview_3_skipframe_turn_off_fen 0: Frame valid remains on during skipped frame. 1: Frame valid output is off during skipped frame.
	6	0x0001	seq_preview_3_skipframe_skip_state 0: No skip state 1: Skip state
	5	0x0000	seq_preview_3_skipframe_skip_led_on Skip first frame when LED On 0: When LED is on, first frame is not skipped. 1: When LED is on, first frame is skipped.
	4:0	X	Reserved
Seq preview 3 skipframe			

Table 28: Driver ID = 2: Auto Exposure Variables

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0002	15:0	0x0000	ae_window_pos (R/W)
	15:8	X	Reserved
	7:4	0x0000	ae_y0 Y0 in units 1/16 of frame height
	3:0	0x0000	ae_x0 X0 in units of 1/16 of frame width
Position of first (upper left) AE window New position written to this variable takes effect only after REFRESH command is given to the sequencer (seq.cmd = 5).			
R0x0003	15:0	0x00FF	ae_window_size (R/W)
	15:8	X	Reserved
	7:4	0x000F	ae_height Height (in units of 1/16 of frame height) decremented by 1
	3:0	0x000F	ae_width Width (in units of 1/16 of frame width) decremented by 1
Size of 4 x 4 array of AE windows New data written to this variable take effect only after REFRESH command is given to the sequencer (seq.cmd = 5).			



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Table 28: Driver ID = 2: Auto Exposure Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0007	15:0	0x0004	ae_gate (R/W)
	AE sensitivity. Changes take effect immediately		
R0x000B	15:0	0x0000	ae_min_index (R/W)
	Minimum allowed zone number. Changes take effect only after REFRESH command.		
R0x000C	15:0	0x0018	ae_max_index (R/W)
	Maximum allowed zone number (that is maximum integration time). Changes take effect only after REFRESH command.		
R0x000D	15:0	0x0010	ae_min_virt_gain (R/W)
	Minimum allowed virtual gain (control analog gains only). Gain is divided by 0x0020; so a value of 0x0020 corresponds to a gain of "1". Changes take effect only after REFRESH command.		
R0x000E	15:0	0x0080	ae_max_virt_gain (R/W)
	Maximum allowed virtual gain (Control analog gains only). Gain is divided by 0x0020; so a value of 0x0020 corresponds to a gain of "1". Changes take effect only after REFRESH command.		
R0x0017	15:0	0x0000	ae_status (R/W)
	15:3	X	Reserved
	2	0x0000	ae_status_ready 0: AE not ready 1: AE ready
	1	0x0000	ae_status_r9_changed Indication of R9 register status change, so a frame should be skipped. 0: R9 register has not changed. 1: R9 register has changed; skip a frame.
	0	0x0000	ae_status_ae_at_limit 0: Indicates the gain or integration has not reached its limit. 1: Indicates the gain or integration has reached its limit.
	AE status		
R0x0018	15:0	0x004B	ae_current_y (RO)
	Last measured luminance		
R0x0019	15:0	0x0279	ae_r12 (RO)
	Current shutter delay value		
R0x001B	15:0	0x0004	ae_index (RO)
	Current zone (integration time)		
R0x001C	15:0	0x0010	ae_virt_gain (RO)
	Current virtual gain		
R0x0022	15:0	0x0010	ae_r9 (R/W)
	Current integration time value		
R0x002D	15:0	0x009D	ae_r9_step (R/W)
	Integration time of one zone		
R0x004A	15:0	0x0032	ae_target_min (R/W)
	Minimum value for the target		
R0x004B	15:0	0x0096	ae_target_max (R/W)
	Maximum value for target		
R0x004C	15:0	0x000C	ae_target_buffer_speed (R/W)
	Maximum allowed change. Setting of 0x0020 results in the fastest AE speed while setting of 0x0001 results in the slowest AE speed.		



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Table 28: Driver ID = 2: Auto Exposure Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x004F	15:0	0x0036	ae_base_target (R/W)
	Base target for exposure. This can be used to change the overall brightness.		

Table 29: Driver ID = 3: Auto White Balance Variables

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0002	15:0	0x0000	awb_window_pos (R/W)
	15:8	X	Reserved
	7:4	0x0000	awb_window_pos_y0 Y0 in units 1/16 of frame height
	3:0	0x0000	awb_window_pos_x0 X0 in units of 1/16 of frame width
	Position of upper left corner of AWB window. New position written to this variable takes effect only after REFRESH command is given to the sequencer (seq.cmd = 5).		
R0x0003	15:0	0x00EF	awb_window_size (R/W)
	15:8	X	Reserved
	7:4	0x000E	awb_window_size_height Height (in units of 1/16 of frame height) decremented by 1.
	3:0	0x000F	awb_window_size_width Width (in units of 1/16 of frame width) decremented by 1.
	Size of AWB window. New size written to this variable takes effect only after REFRESH command is given to the sequencer (seq.cmd = 5).		
R0x0006	15:0	0x01B3	awb_ccm_l_0 (R/W)
	Left color correction matrix element K11. Value is encoded in signed two's complement form; 0x0100 means 1.0. Changes take effect immediately.		
R0x0008	15:0	0xFF1A	awb_ccm_l_1 (R/W)
	Left color correction matrix element K12. Value is encoded in signed two's complement form; 0x0100 means 1.0. Changes take effect immediately.		
R0x000A	15:0	0x0033	awb_ccm_l_2 (R/W)
	Left color correction matrix element K13. Value is encoded in signed two's complement form; 0x0100 means 1.0. Changes take effect immediately.		
R0x000C	15:0	0xFF8D	awb_ccm_l_3 (R/W)
	Left color correction matrix element K21. Value is encoded in signed two's complement form; 0x0100 means 1.0. Changes take effect immediately.		
R0x000E	15:0	0x0200	awb_ccm_l_4 (R/W)
	Left color correction matrix element K22. Value is encoded in signed two's complement form; 0x0100 means 1.0. Changes take effect immediately.		
R0x0010	15:0	0xFF73	awb_ccm_l_5 (R/W)
	Left color correction matrix element K23. Value is encoded in signed two's complement form; 0x0100 means 1.0. Changes take effect immediately.		
R0x0012	15:0	0xFF9A	awb_ccm_l_6 (R/W)
	Left color correction matrix element K31. Value is encoded in signed two's complement form; 0x0100 means 1.0. Changes take effect immediately.		



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Table 29: Driver ID = 3: Auto White Balance Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0014	15:0	0xFF1A	awb_ccm_l_7 (R/W)
			Left color correction matrix element K32. Value is encoded in signed two's complement form; 0x0100 means 1.0. Changes take effect immediately.
R0x0016	15:0	0x024D	awb_ccm_l_8 (R/W)
			Left color correction matrix element K33. Value is encoded in signed two's complement form; 0x0100 means 1.0. Changes take effect immediately.
R0x0018	15:0	0x001C	awb_ccm_l_9 (R/W)
			Left color correction matrix red/green gain ratio. Value is interpreted as unsigned; 0x0020 means 1.0. Changes take effect immediately.
R0x001A	15:0	0x003A	awb_ccm_l_10 (R/W)
			Left color correction matrix blue/green gain ratio. Value is interpreted as unsigned; 0x0020 means 1.0. Changes take effect immediately.
R0x001C	15:0	0x000D	awb_ccm_rl_0 (R/W)
			Delta color correction matrix element D11. Value is encoded in signed two's complement form; 0x0100 means 1.0. Changes take effect immediately.
R0x001E	15:0	0x0033	awb_ccm_rl_1 (R/W)
			Delta color correction matrix element D12. Value is encoded in signed two's complement form; 0x0100 means 1.0. Changes take effect immediately.
R0x0020	15:0	0xFFDA	awb_ccm_rl_2 (R/W)
			Delta color correction matrix element D13. Value is encoded in signed two's complement form; 0x0100 means 1.0. Changes take effect immediately.
R0x0022	15:0	0x000D	awb_ccm_rl_3 (R/W)
			Delta color correction matrix element D21. Value is encoded in signed two's complement form; 0x0100 means 1.0. Changes take effect immediately.
R0x0024	15:0	0xFFCD	awb_ccm_rl_4 (R/W)
			Delta color correction matrix element D22. Value is encoded in signed two's complement form; 0x0100 means 1.0. Changes take effect immediately.
R0x0026	15:0	0x0026	awb_ccm_rl_5 (R/W)
			Delta color correction matrix element D23. Value is encoded in signed two's complement form; 0x0100 means 1.0. Changes take effect immediately.
R0x0028	15:0	0x004D	awb_ccm_rl_6 (R/W)
			Delta color correction matrix element D31. Value is encoded in signed two's complement form; 0x0100 means 1.0. Changes take effect immediately.
R0x002A	15:0	0x001A	awb_ccm_rl_7 (R/W)
			Delta color correction matrix element D32. Value is encoded in signed two's complement form; 0x0100 means 1.0. Changes take effect immediately.
R0x002C	15:0	0xFF9A	awb_ccm_rl_8 (R/W)
			Delta color correction matrix element D33. Value is encoded in signed two's complement form; 0x0100 means 1.0. Changes take effect immediately.
R0x002E	15:0	0x000C	awb_ccm_rl_9 (R/W)
			Delta color correction matrix red/green gain ratio. Value is interpreted as unsigned; 0x0020 means 1.0. Changes take effect immediately.
R0x0030	15:0	0xFFEC	awb_ccm_rl_10 (R/W)
			Delta color correction matrix blue/green gain ratio. Value is interpreted as unsigned; 0x0020 means 1.0. Changes take effect immediately.



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Table 29: Driver ID = 3: Auto White Balance Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0032	15:0	0x01BA	awb_ccm_0 (R/W)
			Current color correction matrix element C11. Value is stored in signed two's complement form; 0x0100 means 1.0.
R0x0034	15:0	0xFF5B	awb_ccm_1 (R/W)
			Current color correction matrix element C12
R0x0036	15:0	0xFFF1	awb_ccm_2 (R/W)
			Current color correction matrix element C13
R0x0038	15:0	0xFFC7	awb_ccm_3 (R/W)
			Current color correction matrix element C21
R0x003A	15:0	0x01B9	awb_ccm_4 (R/W)
			Current color correction matrix element C22
R0x003C	15:0	0xFF87	awb_ccm_5 (R/W)
			Current color correction matrix element C23
R0x003E	15:0	0xFFF9	awb_ccm_6 (R/W)
			Current color correction matrix element C31
R0x0040	15:0	0xFF32	awb_ccm_7 (R/W)
			Current color correction matrix element C32
R0x0042	15:0	0x01DC	awb_ccm_8 (R/W)
			Current color correction matrix element C33
R0x0044	15:0	0x003C	awb_ccm_9 (R/W)
			Current color correction matrix red/green gain ratio. Value is interpreted as unsigned; 0x0020 means 1.0.
R0x0046	15:0	0x002B	awb_ccm_10 (R/W)
			Current color correction matrix blue/green gain ratio. Value is interpreted as unsigned; 0x0020 means 1.0.
R0x0048	15:0	0x0008	awb_gain_buffer_speed (R/W)
			Speed of AWB digital gains buffering (0x0020 = fastest, 0x0001 = slowest).
R0x0049	15:0	0x0002	awb_jump_divisor (R/W)
			Number of steps in which AWB driver is expected to reach color target. The smaller the number, the faster the AWB reaches the color target.
R0x004A	15:0	0x0059	awb_gain_min_r (R/W)
			Minimum allowed AWB digital Red gain (0x0080 means 1.0) Changes take effect immediately.
R0x004B	15:0	0x00A6	awb_gain_max_r (R/W)
			Maximum allowed AWB digital Red gain (0x0080 means 1.0). Changes take effect immediately.
R0x004C	15:0	0x0059	awb_gain_min_b (R/W)
			Minimum allowed AWB digital Blue gain (0x0080 means 1.0) Changes take effect immediately.
R0x004D	15:0	0x00A6	awb_gain_max_b (R/W)
			Maximum allowed AWB digital Blue gain (0x0080 means 1.0). Changes take effect immediately.
R0x004E	15:0	0x0080	awb_gain_r (R/W)
			Current R digital gain (0x0080 means 1.0)
R0x004F	15:0	0x0080	awb_gain_g (R/W)
			Current G digital gain (0x0080 means 1.0)
R0x0050	15:0	0x0080	awb_gain_b (R/W)
			Current B digital gain (0x0080 means 1.0)
R0x0051	15:0	0x0000	awb_ccm_position_min (R/W)
			Leftmost position of color correction matrix (corresponding to incandescent light). Changes effect immediately.



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Table 29: Driver ID = 3: Auto White Balance Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0052	15:0	0x007F	awb_ccm_position_max (R/W)
	Rightmost position of color correction matrix (corresponding to daylight). Changes take effect immediately.		
R0x0053	15:0	0x0040	awb_ccm_position (R/W)
	Current position of color correction matrix (0x0000 corresponds to incandescent light, 0x007F to daylight). Changes take effect immediately.		
R0x0054	15:0	0x0080	awb_saturation (R/W)
	Saturation of color correction matrix (0x0080 means 100%). Changes take effect immediately.		
R0x0055	15:0	0x0000	awb_mode (R/W)
	15:7	X	Reserved
	6	0x0000	awb_mode_normccm_off CCM Normalization 0: Enable CCM normalization. 1: Disable CCM normalization.
	5	0x0000	awb_mode_force_unit_dgains Force AWB digital gains to unity 0: AWB digital gains are not locked to unity. 1: AWB digital gains are locked to unity; program value into awb.CCMPosition to manually set matrix position.
	4:0	0x0000	Reserved
	AWB mode		
R0x0056	15:0	0x0000	awb_gain_r_buf (R/W)
	Time-buffered R gain (0x1000 means 1.0)		
R0x0058	15:0	0x0000	awb_gain_b_buf (R/W)
	Time-buffered B gain (0x1000 means 1.0)		
R0x005D	15:0	0x0078	awb_steady_b_gain_out_min (R/W)
	Blue gain minimum threshold to start search. When AWB blue digital gain falls below this threshold, the AWB driver starts searching for new color correction matrix position. Threshold setting of 0x0080 corresponds to a gain of 1.0; higher setting means higher gain. Changes take effect immediately.		
R0x005E	15:0	0x0086	awb_steady_b_gain_out_max (R/W)
	Blue gain maximum threshold to start search. When AWB blue digital gain goes above this threshold, the AWB driver starts searching for new color correction matrix position. Threshold setting of 0x0080 corresponds to a gain of 1.0; higher setting means higher gain. Changes take effect immediately.		
R0x005F	15:0	0x007E	awb_steady_b_gain_in_min (R/W)
	Blue gain minimum threshold to end search. When AWB blue digital gain is between awb.steadyBGainInMin and awb.steadyBGainInMax, the AWB driver maintains current color correction matrix position. Threshold setting of 0x0080 corresponds to a gain of 1.0; higher setting means higher gain. Changes take effect immediately.		
R0x0060	15:0	0x0082	awb_steady_b_gain_in_max (R/W)
	Blue gain maximum threshold to end search. See awb.steadyBGainInMin above. Changes take effect immediately.		
R0x0065	15:0	0x0010	awb_x0 (R/W)
	CCM position threshold between day and incandescent normalization coefficients. Changes take effect immediately.		
R0x0066	15:0	0x00AA	awb_kr_l (R/W)
	CCM red row normalization coefficient for left matrix position (0x0080 – minimal number). Changes take effect immediately.		



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Table 29: Driver ID = 3: Auto White Balance Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0067	15:0	0x0096	awb_kg_l (R/W)
	CCM green row normalization coefficient for left matrix position (0x0080 – minimal number). Changes take effect immediately.		
R0x0068	15:0	0x0080	awb_kb_l (R/W)
	CCM blue row normalization coefficient for left matrix position (0x0080 – minimal number). Changes take effect immediately.		
R0x0069	15:0	0x0091	awb_kr_r (R/W)
	CCM red row normalization coefficient for right matrix position (0x0080 – minimal number). Changes take effect immediately.		
R0x006A	15:0	0x0082	awb_kg_r (R/W)
	CCM green row normalization coefficient for right matrix position (0x0080 – minimal number). Changes take effect immediately.		
R0x006B	15:0	0x0082	awb_kb_r (R/W)
	CCM blue row normalization coefficient for right matrix position (0x0080 – minimal number). Changes take effect immediately.		

Table 30: Driver ID = 4: Anti-Flicker Variables

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0002	15:0	0x001D	fd_window_posh (R/W)
	15:8	X	Reserved
	7:4	0x0001	fd_width_x0 Left boundary of the flicker measurement window (in units of 1/16 of frame width)
	3:0	0x000D	fd_width Width (in units of 1/16 of frame width) decremented by 1
	Width of flicker measurement window and position of its left boundary, X0		
R0x0003	15:0	0x0004	fd_window_height (R/W)
	15:6	X	Reserved
	5:0	0x0004	fd_window_rows Flicker measurement window height in rows
	Flicker measurement window height		



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Table 30: Driver ID = 4: Anti-Flicker Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0004	15:0	0x0000	fd_mode (R/W)
	15:8	X	Reserved
	7	0x0000	fd_mode_manual_mode Manual flicker mode enable 0: Manual mode disabled 1: Manual mode enabled
	6	0x0000	fd_mode_curr_settings Manual flicker mode selection 0: 60Hz settings 1: 50Hz settings
	5	RO	fd_mode_curr_flicker_state Indicates current flicker detection settings 0: 60Hz 1: 50Hz
	4	0x0000	Reserved
	3:0	RO	Reserved
	Flicked detection switches and indicators.		
R0x0008	15:0	0x001E	fd_search_f1_50 (R/W)
	Lower limit of period range of interest in 50Hz flicker detection. If the FD driver is searching for 50Hz flicker and detects in a frame a flicker-like signal whose period in rows is between fl.search_f1_50 and fl.search_f2_50, it counts the frame as one containing flicker.		
R0x0009	15:0	0x0020	fd_search_f2_50 (R/W)
	Upper limit of period range of interest in 50Hz flicker detection. See fd.search_f1_50 above.		
R0x000A	15:0	0x0025	fd_search_f1_60 (R/W)
	Lower limit of period range of interest in 60Hz flicker detection. If the FD driver is searching for 60Hz flicker and detects in a frame a flicker-like signal whose period in rows is between fl.search_f1_60 and fl.search_f2_60, it counts the frame as one containing flicker.		
R0x000B	15:0	0x0027	fd_search_f2_60 (R/W)
	Upper limit of period range of interest in 60Hz flicker detection. See fd.search_f1_60 above.		
R0x000D	15:0	0x0003	fd_stat_min (R/W)
	Flicker is considered detected if fd.stat_min out of fd.stat_max consecutive frames contain flicker (periodic signal of sought frequency).		
R0x000E	15:0	0x0005	fd_stat_max (R/W)
	See fd.stat_min.		
R0x0010	15:0	0x0005	fd_min_amplitude (R/W)
	Signals less than this threshold are ignored.		
R0x0011	15:0	0x009D	fd_r9_step_f60_a (R/W)
	Minimal shutter width step for 60Hz AC in context A		
R0x0013	15:0	0x00BC	fd_r9_step_f50_a (R/W)
	Minimal shutter width step for 50Hz AC in context A		
R0x0015	15:0	0x0000	fd_r9_step_f60_b (R/W)
	Minimal shutter width step for 60Hz AC in context B		
R0x0017	15:0	0x00E0	fd_r9_step_f50_b (R/W)
	Minimal shutter width step for 50Hz AC in context B		



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Table 31: Driver ID = 7: Mode Variables

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0003	15:0	0x0280	mode_output_width_a (R/W)
	Output size of final image for context A (preview). Must be equal to or smaller than crop dimension. Changes take effect only after REFRESH command.		
R0x0005	15:0	0x0200	mode_output_height_a (R/W)
	Output size of final image for context A (preview). Must be equal to or smaller than crop dimension. Changes take effect only after REFRESH command.		
R0x0007	15:0	0x0500	mode_output_width_b (R/W)
	Output size of final image for context B (snapshot/video). Must be equal to or smaller than crop dimension. Changes take effect only after REFRESH command.		
R0x0009	15:0	0x0400	mode_output_height_b (R/W)
	Output size of final image for context B (snapshot/video). Must be equal to or smaller than crop dimension. Changes take effect only after REFRESH command.		
R0x000D	15:0	0x0000	mode_sensor_row_start_a (R/W)
	The first row of visible pixels to be read out (not counting any dark rows that may be read) in context A (preview). To move the image window, set this register to the starting "Y" value. Must be an even value. Changes take effect only after REFRESH_MODE command.		
R0x000F	15:0	0x0000	mode_sensor_col_start_a (R/W)
	The first column of visible pixels to be read out (not counting any dark columns that may be read) in context A (preview). To move the image window, set this register to the starting "X" value. Must be an even value. Changes take effect only after REFRESH_MODE command.		
R0x0011	15:0	0x04BD	mode_sensor_row_end_a (R/W)
	The last row of visible pixels to be read out. Must be an odd value in context A (preview). Changes take effect only after REFRESH_MODE command.		
R0x0013	15:0	0x064D	mode_sensor_col_end_a (R/W)
	The last column of visible pixels to be read out in context A (preview). Must be an odd value. Changes take effect only after REFRESH_MODE command.		
R0x0015	15:0	0x2112	mode_sensor_row_speed_a (R/W)
	15:3	X	Reserved
	2:0	0x0002	mode_sensor_row_speed_a_pix_clk_speed A programmed value of N gives a pixel clock period of N system clocks. A value of "0" is illegal; it causes the clock to stop.
	Context A (preview) row speed. Changes take effect only after REFRESH_MODE command.		



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Register Description

Table 31: Driver ID = 7: Mode Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0017	15:0	0x046C	mode_sensor_read_mode_a (R/W)
	15:12	X	Reserved
	11	0x0000	mode_sensor_read_mode_a_x_bin_en Enable analog binning in X (column) direction. When set, x_odd_inc must be set to 0x0003 and y_odd_inc must be set to 0x0001.
	10	0x0001	mode_sensor_read_mode_a_xy_bin_en Enable analog binning in X and Y (column and row) directions. When set, x_odd_inc and y_odd_inc must be set to 0x0003.
	9	0x0000	mode_sensor_read_mode_a_low_power_enable Low power mode enable 0: Low power mode is disabled. 1: Low power mode is enabled.
	8	X	Reserved
	7:5	0x0003	mode_sensor_read_mode_a_x_odd_inc Increment applied to odd addresses in X (column) direction. 0x0001: Normal readout. 0x0003: Read out alternate pixel pairs to halve the amount of horizontal data in a frame.
	4:2	0x0003	mode_sensor_read_mode_a_y_odd_inc Increment applied to odd addresses in Y (row) direction. 0x0001: Normal readout. 0x0003: Read out alternate pixel pairs to halve the amount of vertical data in a frame.
	1	0x0000	mode_sensor_read_mode_a_vert_flip 0: Normal readout . 1: Readout is flipped (mirrored) vertically so that the row specified by sensor_row_end_a is read out of the sensor first.
	0	0x0000	mode_sensor_read_mode_a_horiz_mirror 0: Normal readout. 1: Readout is mirrored horizontally so that the column specified by sensor_col_end_a is read out of the sensor first.
Context A (preview) read mode register. Changes take effect only after REFRESH_MODE command.			
R0x0019	15:0	0x007B	mode_sensor_fine_correction_a (R/W)
	Context A (preview) fine integration time correction factor. This is an offset that is applied to the programmed value of fine_integration_time such that the actual integration time matches the integration time equation. Changes take effect only after REFRESH_MODE command.		
R0x001B	15:0	0x0408	mode_sensor_fine_it_min_a (R/W)
	Context A (preview) minimal fine integration time. Changes take effect only after REFRESH_MODE command.		
R0x001D	15:0	0x00AB	mode_sensor_fine_it_max_margin_a (R/W)
	Context A (preview) maximal fine integration time margin. Changes take effect only after REFRESH_MODE command.		
R0x001F	15:0	0x0293	mode_sensor_frame_length_a (R/W)
	The number of complete lines (rows) in the output frame in context A (preview). This includes visible lines and vertical blanking lines. Changes take effect only after REFRESH_MODE command.		
R0x0021	15:0	0x07D0	mode_sensor_line_length_pck_a (R/W)
	The number of pixel clock periods in one line (row) time in context A (preview). This includes visible pixels and horizontal blanking time. Changes take effect only after REFRESH_MODE command.		



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Table 31: Driver ID = 7: Mode Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0023	15:0	0x0004	mode_sensor_row_start_b (R/W)
	The first row of visible pixels to be read out (not counting any dark rows that may be read) in context B (snapshot/video). To move the image window, set this register to the starting "Y" value. Must be an even value. Changes take effect only after REFRESH_MODE command.		
R0x0025	15:0	0x0004	mode_sensor_col_start_b (R/W)
	The first column of visible pixels to be read out (not counting any dark columns that may be read) in context B (snapshot/video). To move the image window, set this register to the starting "X" value. Must be an even value. Changes take effect only after REFRESH_MODE command.		
R0x0027	15:0	0x04BB	mode_sensor_row_end_b (R/W)
	The last row of visible pixels to be read out. Must be an odd value in context B (snapshot/video). Changes take effect only after REFRESH_MODE command.		
R0x0029	15:0	0x064B	mode_sensor_col_end_b (R/W)
	The last column of visible pixels to be read out in context B (snapshot/video). Must be an odd value. Changes take effect only after REFRESH_MODE command.		
R0x002B	15:0	0x2111	mode_sensor_row_speed_b (R/W)
	15:3	X	Reserved
	2:0	0x0001	mode_sensor_row_speed_b_pix_clk_speed A programmed value of N gives a pixel clock period of N system clocks. A value of "0" is illegal; it causes the clock to stop.
	Context B (snapshot/video) row speed. Changes take effect only after REFRESH_MODE command.		
R0x002D	15:0	0x0024	mode_sensor_read_mode_b (R/W)
	15:12	X	Reserved
	11	0x0000	mode_sensor_read_mode_b_x_bin_en Enable analog binning in X (column) direction. When set, x_odd_inc must be set to 0x0003 and y_odd_inc must be set to 0x0001.
	10	0x0000	mode_sensor_read_mode_b_xy_bin_en Enable analog binning in X and Y (column and row) directions. When set, x_odd_inc and y_odd_inc must be set to 0x0003.
	9	0x0000	mode_sensor_read_mode_b_low_power_mode
	8	X	Reserved
	7:5	0x0001	mode_sensor_read_mode_b_x_odd_inc Increment applied to odd addresses in X (column) direction. 0x0001: Normal readout. 0x0003: Read out alternate pixel pairs to halve the amount of horizontal data in a frame.
	4:2	0x0001	mode_sensor_read_mode_b_y_odd_inc Increment applied to odd addresses in Y (row) direction. 0x0001: Normal readout. 0x0003: Read out alternate pixel pairs to halve the amount of vertical data in a frame.
	1	0x0000	mode_sensor_read_mode_b_vert_flip 0x0000: Normal readout. 0x0001: Readout is flipped (mirrored) vertically so that the row specified by sensor_row_end_b is read out of the sensor first.
	0	0x0000	mode_sensor_read_mode_b_horiz_mirror 0: Normal readout 1: Readout is mirrored horizontally so that the column specified by sensor_col_end_b is read out of the sensor first.
	Context B (snapshot/video) read mode register. Changes take effect only after REFRESH_MODE command.		



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Register Description

Table 31: Driver ID = 7: Mode Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x002F	15:0	0x00A4	mode_sensor_fine_correction_b (R/W)
			Context B (snapshot/video) fine integration time correction factor. This is an offset that is applied to the programmed value of fine_integration_time such that the actual integration time matches the integration time equation. Changes take effect only after REFRESH_MODE command.
R0x0031	15:0	0x0408	mode_sensor_fine_it_min_b (R/W)
			Context B (snapshot/video) minimal fine integration time. Changes take effect only after REFRESH_MODE command.
R0x0033	15:0	0x00A4	mode_sensor_fine_it_max_margin_b (R/W)
			Context B (snapshot/video) maximal fine integration time margin. Changes take effect only after REFRESH_MODE command.
R0x0035	15:0	0x04ED	mode_sensor_frame_length_b (R/W)
			The number of complete lines (rows) in the output frame in context B (snapshot/video). This includes visible lines and vertical blanking lines. Changes take effect only after REFRESH_MODE command.
R0x0037	15:0	0x0D06	mode_sensor_line_length_pck_b (R/W)
			The number of pixel clock periods in one line (row) time in context B (snapshot/video). This includes visible pixels and horizontal blanking time. Changes take effect only after REFRESH_MODE command.
R0x0039	15:0	0x0000	mode_crop_x0_a (R/W)
			Lower-x decimator zoom window for context A (preview). Changes take effect only after REFRESH command.
R0x003B	15:0	0x027F	mode_crop_x1_a (R/W)
			Upper-x decimator zoom window for context A (preview). Changes take effect only after REFRESH command.
R0x003D	15:0	0x0000	mode_crop_y0_a (R/W)
			Lower-y decimator zoom window for context A (preview). Changes take effect only after REFRESH command.
R0x003F	15:0	0x01FF	mode_crop_y1_a (R/W)
			Upper-y decimator zoom window for context A (preview). Changes take effect only after REFRESH command.
R0x0047	15:0	0x0000	mode_crop_x0_b (R/W)
			Lower-x decimator zoom window for context B (snapshot/video). Changes take effect only after REFRESH command.
R0x0049	15:0	0x04FF	mode_crop_x1_b (R/W)
			Upper-x decimator zoom window for context B (snapshot/video). Changes take effect only after REFRESH command.
R0x004B	15:0	0x0000	mode_crop_y0_b (R/W)
			Lower-y decimator zoom window for context B (snapshot/video). Changes take effect only after REFRESH command.
R0x004D	15:0	0x03FF	mode_crop_y1_b (R/W)
			Upper-y decimator zoom window for context B (snapshot/video). Changes take effect only after REFRESH command.



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Table 31: Driver ID = 7: Mode Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0055	15:0	0x0000	mode_output_format_a (R/W)
	15:9	X	Reserved
	8	0x0000	mode_output_format_a_config_processed_bayer Turn on processed Bayer output. Chip output data in Bayer format. As a result data rate decreases by a factor of two.
	7:6	0x0000	mode_output_format_a_rgb_format RGB output format: 0x0000: 16-bit RGB565 0x0001: 15-bit RGB555 0x0002: 12-bit RGB444x 0x0003: 12-bit RGBx444
	5	0x0000	mode_output_format_a_output_mode RGB/YUV Output 0: Output YUV 1: Output RGB (see mode_output_format_a_output_mode[7:6] for detail).
	4	0x0000	Reserved
	3	0x0000	mode_output_format_a_monochrome This bit forces the output to produce a monochrome output. 0: Monochrome output is disabled. 1: Monochrome output is enabled.
	2	0x0000	Reserved
	1	0x0000	mode_output_format_a_swap_chrominance_luma Chrominance swap. This bit is subject to synchronous update. 0: Normal mode. 1: In YUV mode, swaps chrominance byte with luminance byte. In RGB mode, swaps odd and even bytes.
	0	0x0000	mode_output_format_a_swap_channels Channel swap for YUV and RGB output. This bit is subject to synchronous update. 0: Normal mode. 1: In YUV output mode, swaps Cb and Cr channels. In RGB mode, swaps R and B.
	Output format configuration for context A (preview). Changes take effect only after REFRESH command.		



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Register Description

Table 31: Driver ID = 7: Mode Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0057	15:0	0x0000	mode_output_format_b (R/W)
	15:9	X	Reserved
	8	0x0000	mode_output_format_b_config_processed_bayer Turn on processed Bayer output. Chip outputs data in Bayer format. As a result data rate decreases by a factor of two.
	7:6	0x0000	mode_output_format_b_rgb_format RGB output format: 0x0000: 16-bit RGB565 0x0001: 15-bit RGB555 0x0002: 12-bit RGB444x 0x0003: 12-bit RGBx444
	5	0x0000	mode_output_format_b_output_mode RGB/YUV Output. 0: Output YUV. 1: Output RGB (see mode_output_format_a_output_mode[7:6] for detail).
	4	0x0000	Reserved
	3	0x0000	mode_output_format_b_monochrome This bit forces the output to produce a monochrome output. 0: Monochrome output is disabled. 1: Monochrome output is enabled.
	2	0x0000	Reserved
	1	0x0000	mode_output_format_b_swap_chrominance_luma Chrominance swap. This bit is subject to synchronous update. 0: Normal mode. 1: In YUV mode, swaps chrominance byte with luminance byte. In RGB mode, swaps odd and even bytes.
	0	0x0000	mode_output_format_b_swap_channels Channel swap for YUV and RGB output. This bit is subject to synchronous update. 0: Normal mode. 1: In YUV output mode, swaps Cb and Cr channels. In RGB mode, swaps R and B.
	Output format configuration for context B (snapshot/video). Changes take effect only after REFRESH command.		
R0x0059	15:0	0x6440	mode_spec_effects_a (R/W)
	15:8	0x0064	mode_spec_effects_a_solarization_thresh
	7	X	Reserved
	6	0x0001	mode_spec_effects_a_dither_luma Dither Selection 0: Dither in all color channels 1: Dither only in luma channel
	5:3	0x0000	mode_spec_effects_a_dither_bit_width Bit width of dither (valid values 1, 2, 3, and 4; no dither if value = 0, 5, 6, or 7)
	2:0	0x0000	mode_spec_effects_a_selection 0x0000: Disabled 0x0001: Monochrome 0x0002: Sepia 0x0003: Negative 0x0004: Solarization with unmodified UV 0x0005: Solarization with -UV
	Special effects selection for context A (preview). Changes take effect only after REFRESH command.		



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Table 31: Driver ID = 7: Mode Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x005B	15:0	0x6440	mode_spec_effects_b (R/W)
	15:8	0x0064	mode_spec_effects_b_solarization_thresh
	7	X	Reserved
	6	0x0001	mode_spec_effects_b_dither_luma Dither Selection 0: Dither in all color channels 1: Dither only in luma channel
	5:3	0x0000	mode_spec_effects_b_dither_bit_width Bit width of dither (valid values 1, 2, 3, and 4; no dither if value = 0, 5, 6, or 7)
	2:0	0x0000	mode_spec_effects_b_selection 0x0000: Disabled 0x0001: Monochrome 0x0002: Sepia 0x0003: Negative 0x0004: Solarization with unmodified UV 0x0005: Solarization with -UV
Special effects selection for context B (snapshot/video). Changes take effect only after REFRESH command.			
R0x005D	15:0	0x0000	mode_y_rgb_offset_a (R/W)
	Y/RGB offset setting for context A (preview). Changes take effect only after REFRESH command.		
R0x005E	15:0	0x0000	mode_y_rgb_offset_b (R/W)
	Y/RGB offset setting for context B (snapshot/video). Changes take effect only after REFRESH command.		
R0x0063	15:0	0x0000	mode_common_mode_settings_fx_sepia_settings (R/W)
	15:8	0x0000	mode_sepia_cb Magnitude of Cb in 0.7 fixed point.
	7:0	0x0000	mode_sepia_cr Magnitude of Cr in 0.7 fixed point.
	This register specifies the chroma values for the sepia effect. In sepia mode, the chroma values of each pixel are set to this value. By default, this register contains a brownish color, but it can be set to an arbitrary color. Changes take effect only after REFRESH command.		
R0x0066	15:0	0x0000	mode_common_mode_settings_test_mode (R/W)
	Test Pattern Generator. Changes take effect only after REFRESH command. 0x0000: Disabled 0x0001: Solid white 0x0002: Grey ramp 0x0003: Color bar ramp 0x0004: Solid white (color bars) 0x0005: Noise		
R0x0068	15:0	0x0005	mode_min_it_time_limit (R/W)
	Mode min it time limit		



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Table 32: Driver ID = 11: Histogram Variables

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0004	15:0	0x0040	hg_max_dark_level (R/W)
	Limits dark-level variable		
R0x0006	15:0	0x0003	hg_black_clip_percent (R/W)
	Amount of black pixels the dark-level subtraction algorithm will try to clip (in percent * 10). For example: 10 = 1 Percent, 3 = 0.3 Percent.		
R0x0008	15:0	0x0010	hg_dark_level (R/W)
	Makes image darker or brighter based on target black clip pixels.		
R0x001B	15:0	0x0000	hg_brightness_metric (R/W)
	A measure of the brightness of a scene. It increases in value as the scene becomes darker. This measure does not correspond to any lux level.		
R0x001F	15:0	0x0084	hg_llmode (R/W)
	15:8	X	Reserved
	7	0x0001	Reserved
	6:4	X	Reserved
	3:0	0x0004	hg_bm_prescale
	Low light mode controls		
R0x0028	15:0	0x157C	hg_ll_brightness_start (R/W)
	Low light settings start based on hg_brightness_metric (0x1B). When hg_brightness_metric < hg_ll_brightness_start, low light settings are not applied. When hg_brightness_metric >= hg_ll_brightness_start, interpolated low light settings are applied. Note: hg_brightness_metric increases in value as the scene appears darker.		
R0x002A	15:0	0x1B58	hg_ll_brightness_stop (R/W)
	Low light settings start based on hg_brightness_metric (0x1B). When hg_brightness_metric > hg_ll_brightness_stop, low light settings are applied. When hg_brightness_metric <= hg_ll_brightness_stop, interpolated low light settings are applied. Note: hg_brightness_metric increases in value as the scene appears darker.		
R0x0037	15:0	0x0003	hg_gamma_morph_ctrl (R/W)
	15:2	X	Reserved
	1:0	0x0003	hg_enable_gamma_morph 0x0000: Disable gamma morph. 0x0001: Use table A. 0x0002: Use table B. 0x0003: Auto-morph between table A and B based on brightness metric.
	Hg gamma morph ctrl		
R0x0038	15:0	0x157C	hg_gamma_start_morph (R/W)
	Starting point for gamma morphing. When the hg_brightness_metric (0x1B) is < hg_gamma_start_morph, gamma table A is used. When the hg_brightness_metric is >= hg_gamma_start_morph, an interpolation of gamma tables A and B is used. Note: hg_brightness_metric increases in value as the scene appears darker.		
R0x003A	15:0	0x1B58	hg_gamma_stop_morph (R/W)
	Ending point for gamma morphing. When the hg_brightness_metric (0x1B) is > hg_gamma_stop_morph, gamma table B is used. When the hg_brightness_metric is <= hg_gamma_stop_morph, an interpolation of gamma tables A and B is used. Note: hg_brightness_metric increases in value as the scene appears darker.		
R0x003C	15:0	0x0000	hg_gamma_table_a_0 (R/W)
	Gamma table A entry for normal light condition		



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Table 32: Driver ID = 11: Histogram Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x003D	15:0	0x001B	hg_gamma_table_a_1 (R/W)
			Gamma table A entry for normal light condition
R0x003E	15:0	0x002E	hg_gamma_table_a_2 (R/W)
			Gamma table A entry for normal light condition
R0x003F	15:0	0x004C	hg_gamma_table_a_3 (R/W)
			Gamma table A entry for normal light condition
R0x0040	15:0	0x0078	hg_gamma_table_a_4 (R/W)
			Gamma table A entry for normal light condition
R0x0041	15:0	0x0098	hg_gamma_table_a_5 (R/W)
			Gamma table A entry for normal light condition
R0x0042	15:0	0x00B0	hg_gamma_table_a_6 (R/W)
			Gamma table A entry for normal light condition
R0x0043	15:0	0x00E8	hg_gamma_table_a_7 (R/W)
			Gamma table A entry for normal light condition
R0x0044	15:0	0x00CF	hg_gamma_table_a_8 (R/W)
			Gamma table A entry for normal light condition
R0x0045	15:0	0x00D9	hg_gamma_table_a_9 (R/W)
			Gamma table A entry for normal light condition
R0x0046	15:0	0x00E1	hg_gamma_table_a_10 (R/W)
			Gamma table A entry for normal light condition
R0x0047	15:0	0x00E8	hg_gamma_table_a_11 (R/W)
			Gamma table A entry for normal light condition
R0x0048	15:0	0x00EE	hg_gamma_table_a_12 (R/W)
			Gamma table A entry for normal light condition
R0x0049	15:0	0x00F2	hg_gamma_table_a_13 (R/W)
			Gamma table A entry for normal light condition
R0x004A	15:0	0x00F6	hg_gamma_table_a_14 (R/W)
			Gamma table A entry for normal light condition
R0x004B	15:0	0x00F9	hg_gamma_table_a_15 (R/W)
			Gamma table A entry for normal light condition
R0x004C	15:0	0x00FB	hg_gamma_table_a_16 (R/W)
			Gamma table A entry for normal light condition
R0x004D	15:0	0x00FD	hg_gamma_table_a_17 (R/W)
			Gamma table A entry for normal light condition
R0x004E	15:0	0x00FF	hg_gamma_table_a_18 (R/W)
			Gamma table A entry for normal light condition
R0x004F	15:0	0x0000	hg_gamma_table_b_0 (R/W)
			Gamma table B entry for low light condition
R0x0050	15:0	0x000F	hg_gamma_table_b_1 (R/W)
			Gamma table B entry for low light condition
R0x0051	15:0	0x001A	hg_gamma_table_b_2 (R/W)
			Gamma table B entry for low light condition
R0x0052	15:0	0x002E	hg_gamma_table_b_3 (R/W)
			Gamma table B entry for low light condition